

Apply filters to SQL queries

Project description

As a security professional at my organization, my job is to investigate security issues to help keep the system secure. I was tasked to examine the organization's data in their employees and log_in_attempts tables to retrieve records from different datasets and investigate the potential security issues.

Retrieve after-hours failed login attempts

The code below demonstrates how I used SQL to query the organization's database to get information about after-hours failed login attempts. Business hours end at 18:00

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_time > '18:00' AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.1	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.14	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.5	0
28	astrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0

The first 2 lines of the screenshot show the command to query all the columns in the log_in_attempts table, the third line shows that it should only return information of failed login attempts that occurred after business hours, which is 18:00

Retrieve login attempts on specific dates

A suspicious event occurred on 2022-05-09. To investigate this event, I had to review all login attempts that occurred on this day and the day before. The code below shows how I performed this task

```

MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address |
| success |
+-----+-----+-----+-----+-----+-----+
| 1 | jrafael | 2022-05-09 | 04:56:27 | CAN | 192.168.243.1 |
40 | 1 |
| 3 | dkot | 2022-05-09 | 06:47:41 | USA | 192.168.151.1 |
62 | 1 |
| 4 | dkot | 2022-05-08 | 02:00:39 | USA | 192.168.178.7 |
1 | 0 |
| 8 | bisles | 2022-05-08 | 01:30:17 | US | 192.168.119.1 |
73 | 0 |

```

The first 2 lines of the screenshot show the command to query all the columns in the log_in_attempts table. On the third line of the code, I used the OR filter to query for login attempts that occurred on 2022-05-08 and 2022-05-09

Retrieve login attempts outside of Mexico

There was suspicious login activity outside of the Mexico branch, so I had to investigate where the login attempts came from. The code below shows how I performed this task.

```

MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT country LIKE 'MEX%';
+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address |
| success |
+-----+-----+-----+-----+-----+-----+
| 1 | jrafael | 2022-05-09 | 04:56:27 | CAN | 192.168.243.1 |
40 | 1 |
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.1 |
2 | 0 |
| 3 | dkot | 2022-05-09 | 06:47:41 | USA | 192.168.151.1 |
62 | 1 |
| 4 | dkot | 2022-05-08 | 02:00:39 | USA | 192.168.178.7 |

```

The first 2 lines of the screenshot show the command to query all the columns in the log_in_attempts table. On the third line of the code, I used the NOT filter to indicate that I don't want information from the Mexico office, and I also used the LIKE filter to specify that even queries containing the abbreviation MEX should not be included in the result.

Retrieve employees in Marketing

The security team needed to perform a security update on employee machines. Only employees in the marketing department who had offices in the East section of the building needed this update. The code below shows how I performed the task of getting the employees' information

```
MariaDB [organization]> SELECT *  
  -> FROM employees  
  -> WHERE department = 'Marketing' AND office LIKE 'East%';  
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+  
|      1000 | a320b137c219 | elarson | Marketing | East-170 |  
|      1052 | a192b174c940 | jdarosa | Marketing | East-195 |  
|      1075 | x573y883z772 | fbautist | Marketing | East-267 |  
|      1088 | k865l965m233 | rgosh | Marketing | East-157 |  
|      1103 | NULL | randerss | Marketing | East-460 |  
|      1156 | a184b775c707 | dellery | Marketing | East-417 |  
|      1163 | h679i515j339 | cwilliam | Marketing | East-216 |  
+-----+-----+-----+-----+-----+  
7 rows in set (0.001 sec)  
  
MariaDB [organization]> #
```

The first 2 lines of the screenshot show the command to query all the columns in the log_in_attempts table. On the third line of the code, I specified that I wanted employees from the Marketing department, then used the AND filter to query only those found in the East section of the building

Retrieve employees in Finance or Sales

The security team needed to perform another security update on the machines of employees in the Finance and Sales departments. The code below shows how I performed this task.

```
MariaDB [organization]> clear  
MariaDB [organization]> SELECT *  
  -> FROM employees  
  -> WHERE department = 'Finance' OR department = 'Sales';  
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+  
|      1003 | d394e816f943 | sgilmore | Finance | South-153 |  
|      1007 | h174i497j413 | wjaffrey | Finance | North-406 |  
|      1008 | i858j583k571 | abernard | Finance | South-170 |  
|      1009 | NULL | lrodriqu | Sales | South-134 |  
|      1010 | k242l212m542 | jlansky | Finance | South-109 |  
|      1011 | l748m120n401 | drosas | Sales | South-292 |  
+-----+-----+-----+-----+-----+
```

The first 2 lines of the screenshot show the command to query all the columns in the log_in_attempts table. On the third line of the code, I used the OR filter to query for information

on the employees table, so I can get the device_id for employees in the Finance or Sales departments.

Retrieve all employees not in IT

The security team needed to make 1 more update to the machines of employees, but this update did not have to include employees in the IT department. The code below shows how I performed this task.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406

The first 2 lines of the screenshot show the command to query all the columns in the log_in_attempts table. On the third line of the code, I used the NOT filter followed by the argument to specify that I wanted information from every department except the IT department

Summary

I used multiple filters to query the organization's database so I could get detailed information about the suspicious login activities and equally filter out information about different departments based on the updates the security team needed to carry out on employee machines.